Name_____ Date_____ Period_____

LESSON 2-10 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Mean or Median?

Maryland Standard 6.SP.B.5d · Learning target – I can choose the mean or the median to describe a data set based on its shape.

REMEMBER

- **Symmetric** data with no outliers: the mean and median are close. Use the **mean**.
- **Skewed** data or data with an **outlier**: the mean is pulled toward the tail. Use the **median**. If the mean and median are far apart, the median is safer.

WORKED EXAMPLE

Eight commutes took 12, 14, 15, 15, 16, 17, 18, and 61 minutes. Which center is better?

Step 1 61 is an outlier, so the data is skewed right.

Step 2 Median: (15 + 16) ÷ 2 = 15.5. Mean: 168 ÷ 8 = 21, greater than 7 of the 8 times.

Answer **The median, 15.5 minutes**

WORD BANK

outlier – a value far from the others
symmetric – both sides look about the same
skewed – a long tail on one side
median – the middle value in order

PRACTICE

show your work · label units

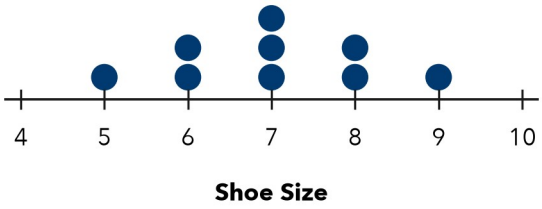
1. Write mean or median for the better measure of center.

Step 1 Look for a value far from the others.

Step 2 No outlier: mean. Outlier: median.

a. 40, 42, 44, 46, 48 ____ **b.** 3, 40, 42, 44, 46, 48 ____ **c.** 8, 9, 10, 11, 12, 13, 75 ____

2. **Shoe Sizes.** The **dot plot** shows the **shoe sizes** of 9 students. Each dot is one student.



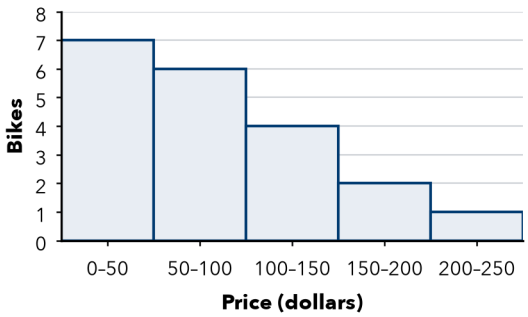
a. Shape: _____

b. Mean: _____

c. Median: _____

d. Better center: _____

3. Used Bikes. The histogram shows the prices of 20 used bikes.



Step 1 Find where the tall bars are and where the tail is.

Step 2 A long tail pulls the mean toward it.

a. Shape: _____

b. Greater, mean or median? _____

c. Better center: _____

4. Allowances. Six friends get weekly allowances of $5, $6, $8, $8, $10, and $41.

Step 1 Mean: add all six, then divide by 6.

Step 2 Median: average the two middle values.

a. Mean: $ ____ **b.** Median: $ ____ **c.** Better center: _____

5. Honest Ads. A shop pays its 5 workers $30,000, $32,000, $34,000, $36,000, and $178,000 a year. Its ad says, "Average pay: $62,000!" Why is this misleading? What center is better?

Frame *The ad is misleading because ___. A better center is ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home: for house prices, is the mean or the median greater? Why?

Name_____ Date_____ Period_____

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Matching Center and Spread

Maryland Standard 6.SP.B.5d · Learning target – I can match a measure of center and a measure of spread to the shape of a data set.

REMEMBER

- Symmetric, no outliers: **mean and MAD**. Skewed or outliers: **median and IQR**.
- Do not mix the pairs. A summary names the shape, center, and spread.

WORKED EXAMPLE

Seven friends sent 3, 4, 5, 6, 7, 8, and 40 texts. Summarize the data.

Step 1 40 is an outlier: use the median and IQR.

Step 2 Median = 6. Q1 = 4, Q3 = 8, IQR = 4

Answer **Median 6 texts, IQR 4 texts**

WORD BANK

mean and MAD – the pair for symmetric data

median and IQR – the pair for skewed data or outliers

IQR – Q3 − Q1

PRACTICE

show your work · label units

1. Write the pair you would report: mean and MAD, or median and IQR.

Step 1 Look for an outlier.

Step 2 No outlier: mean and MAD. Outlier: median and IQR.

a. 15, 16, 17, 18, 19 _____

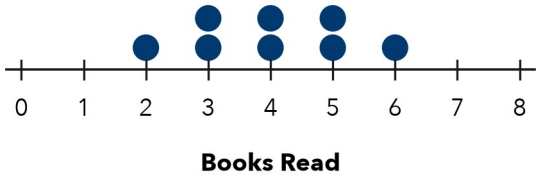
b. 15, 16, 17, 18, 19, 20, 95 _____

2. **Books Read.** Books read by 8 students in a month.

a. Pair to report: _____

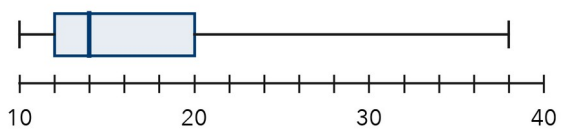
b. Center: _____ books

c. Spread: _____ books



Books Read

3. **Wait Times.** The **box plot** shows deli **wait times** in minutes. Read the median and the quartiles from the box.



   a. Shape: _____

   b. Median: _____ min

   c. IQR: _____ min

4. **Phone Calls.** Seven calls lasted 12, 15, 16, 18, 20, 22, and 70 minutes.

   Step 1 70 is an outlier, so use the median and IQR.

   Step 2 Q1 = middle of the lower 3; Q3 = middle of the upper 3.

   a. Pair: _____ **b.** Median: ___ min **c.** IQR: ___ min

5. **Find the Error.** Kenji says 2, 20, 21, 22, 23, 24, 25 is not skewed because no value is far above the rest. He reports the mean and MAD. Explain his error.

   Frame *Kenji is wrong because 2 is ___. He should report ___.*

TEST PRACTICE

choose the best answer

6. Which pair best describes 21, 23, 24, 25, 26, 27, 80?

   A mean and MAD **B** median and IQR **C** mean and IQR **D** median and MAD

7. A histogram is tall in the middle, with short, equal bars on both ends. Which pair should you report?

   A mean and MAD **B** median and IQR **C** mean and IQR **D** median and MAD

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home: for the prices of all cars on a street, would you report the mean or the median? Why?